Event Triggers

<Practice with exploring event triggers concept>

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About me:

- Java Developer
- Co-Organizer of Codeus community

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What is event trigger?

Event trigger - a special type of trigger that fires on database-level events (DDL operations), not on table data changes.

Office Security Analogy

Regular Triggers

Security cameras in rooms (Monitor what happened in specific rooms)

Event Triggers

Building access control system (Monitor who enters/exits, opens/closes doors)

Regular triggers vs Event triggers

Regular triggers		Event triggers
Respond to data manipulation	Responce	Respond to DB structure changes
Table-specific and localized	Scope	Have database-wide visibility and control
Can fire before, after, or instead of operations	Timing	Fire before or after entire DDL statements
Handle business logic and data validation	Use cases	Auditing, security, and schema governance

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Event Trigger Types

`ddl_command_start` - BEFORE DDL execution

`ddl_command_end` - AFTER DDL execution

`table_rewrite` - When table is rewritten (PostgreSQL)

`sql_drop` - When objects are dropped

DDL commands that fire event triggers

Object Creation

CREATE TABLE, CREATE INDEX, CREATE VIEW, CREATE FUNCTION

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DDL commands that fire event triggers

Object Creation

Object Modification

ALTER TABLE, ALTER INDEX, ALTER VIEW, ALTER FUNCTION

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DDL commands that fire event triggers

Object Creation

Object Modification

Object Deletion

DROP TABLE, DROP INDEX, DROP VIEW, DROP FUNCTION

DDL commands that fire event triggers

Object Creation

Object Modification

Object Deletion

Access Control

GRANT, REVOKE

DDL commands that fire event triggers

Object Creation

Object Modification

Object Deletion

Access Control

Documentation

COMMENT ON TABLE, COMMENT ON COLUMN

When to use event triggers

DDL
Auditing

"Who created/modified/deleted what and when?"

When to use event triggers

DDL Auditing

Security Protection

"Prevent accidental DROP operations in production"

When to use event triggers

DDL Auditing

Security Protection

Automatic Documentation

"Auto-comment new tables with creation info"

When to use event triggers

- DDL Auditing
- Security Protection
- Automatic Documentation
- **Team Notification**

"Notify Slack when schema changes happen"

Basic Event Trigger Example

Step 1: Create the Function

```sql
CREATE OR REPLACE FUNCTION simple_ddl_log()
    RETURNS event_trigger
    LANGUAGE plpgsql
AS $$
BEGIN
    RAISE NOTICE 'DDL executed: % by user %', tg_tag, session_user;
END;
$$;
```

Basic Event Trigger Example

Step 2: Create the Event Trigger

```sql
CREATE EVENT TRIGGER ddl_notifier
    ON ddl_command_start
    EXECUTE FUNCTION simple_ddl_log();
```

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Basic Event Trigger Example

Step 3: Test It

```sql
CREATE TABLE test_table (id SERIAL);
-- Result: NOTICE: DDL executed: CREATE TABLE by user postgres

DROP TABLE test_table;
-- Result: NOTICE: DDL executed: DROP TABLE by user postgres
```

Event Trigger Limitations

What Event Triggers CAN DO

- Read data from tables
- Log events and metadata
- Block operations (via `RAISE EXCEPTION`)
- Get comprehensive DDL operation details

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Event Trigger Limitations

What Event Triggers CANNOT DO

- Modify table data (read-only access)
- Use transaction commands (`COMMIT`, `ROLLBACK`)
- Work with temporary tables
- Fire on system catalog operations

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Best Practices Using Event Triggers

DO

- **Keep functions simple - Event triggers should be fast**
- **Log rather than block - Better to record than prevent**
- **Handle exceptions** - Always use `EXCEPTION` blocks
- **Test thoroughly - Verify on test databases first**
- **Document purpose - Comment why each trigger exists**

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Best Practices Using Event Triggers

DON'T

- Complex logic - Keep trigger functions simple
- Long operations - Avoid time-consuming calculations
- Circular dependencies - Triggers calling themselves
- Data modifications - Event triggers are for monitoring

Thank you

- Author: Yevhenii Fortuna
- My LinkedIn: https://www.linkedin.com/in/yevhenii-fortuna/
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